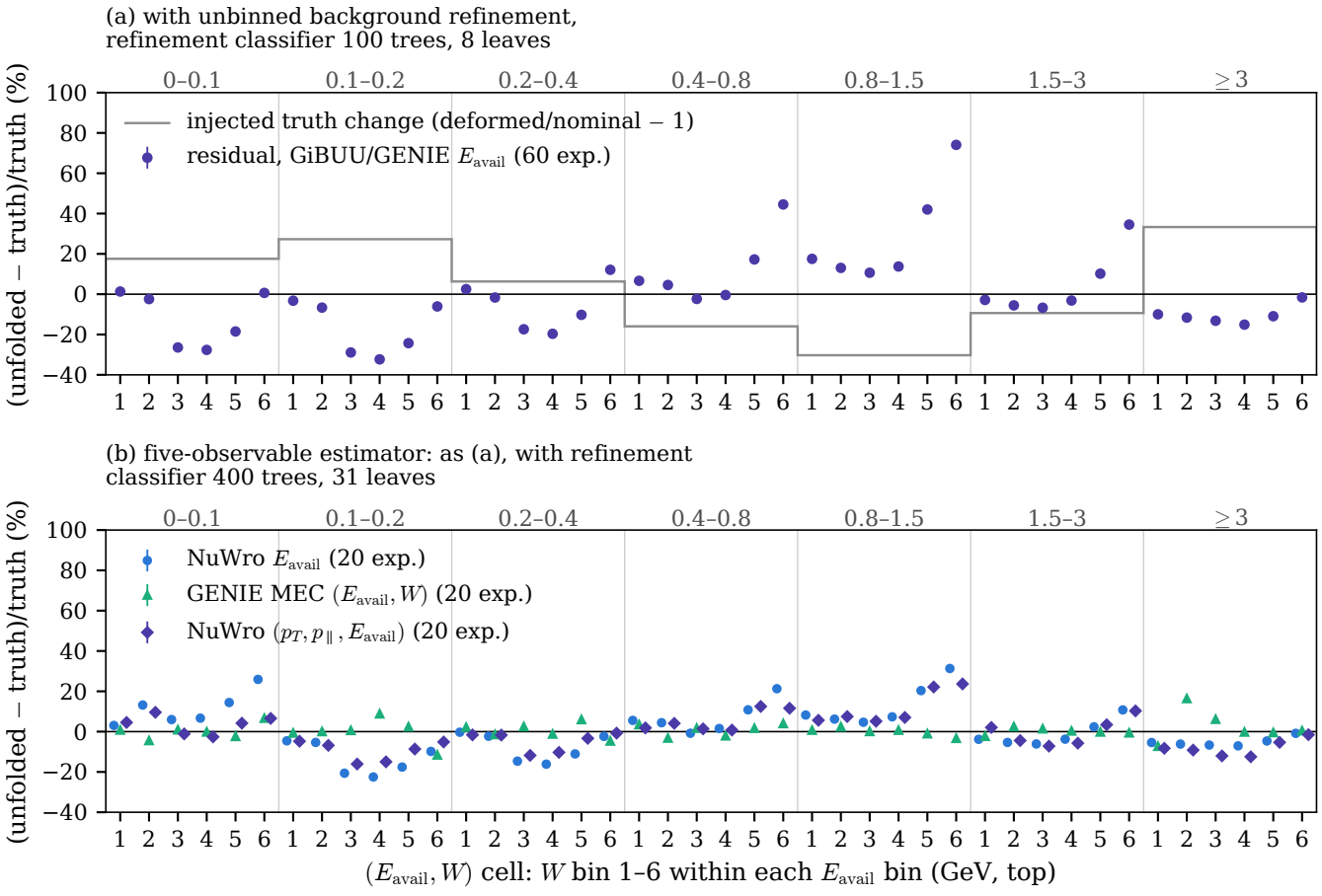
(a) with unbinned background refinement,
refinement classifier 100 trees, 8 leaves
0–0.1
0.1–0.2
0.2–0.4
0.4–0.8
0.8–1.5
1.5–3
≥ 3
injected truth change (deformed/nominal − 1)
residual, GiBUU/GENIE E_{avail} (60 exp.)
(unfolded − truth)/truth (%)
(b) five-observable estimator: as (a), with refinement
classifier 400 trees, 31 leaves
NuWro E_{avail} (20 exp.)
GENIE MEC (E_{avail}, W) (20 exp.)
NuWro $(p_T, p_\parallel, E_{\mathrm{avail}})$ (20 exp.)
(E_{avail}, W) cell: W bin 1–6 within each E_{avail} bin (GeV, top)